



EC2.103

Analog Electronic Circuits

Quadrature Down Converter

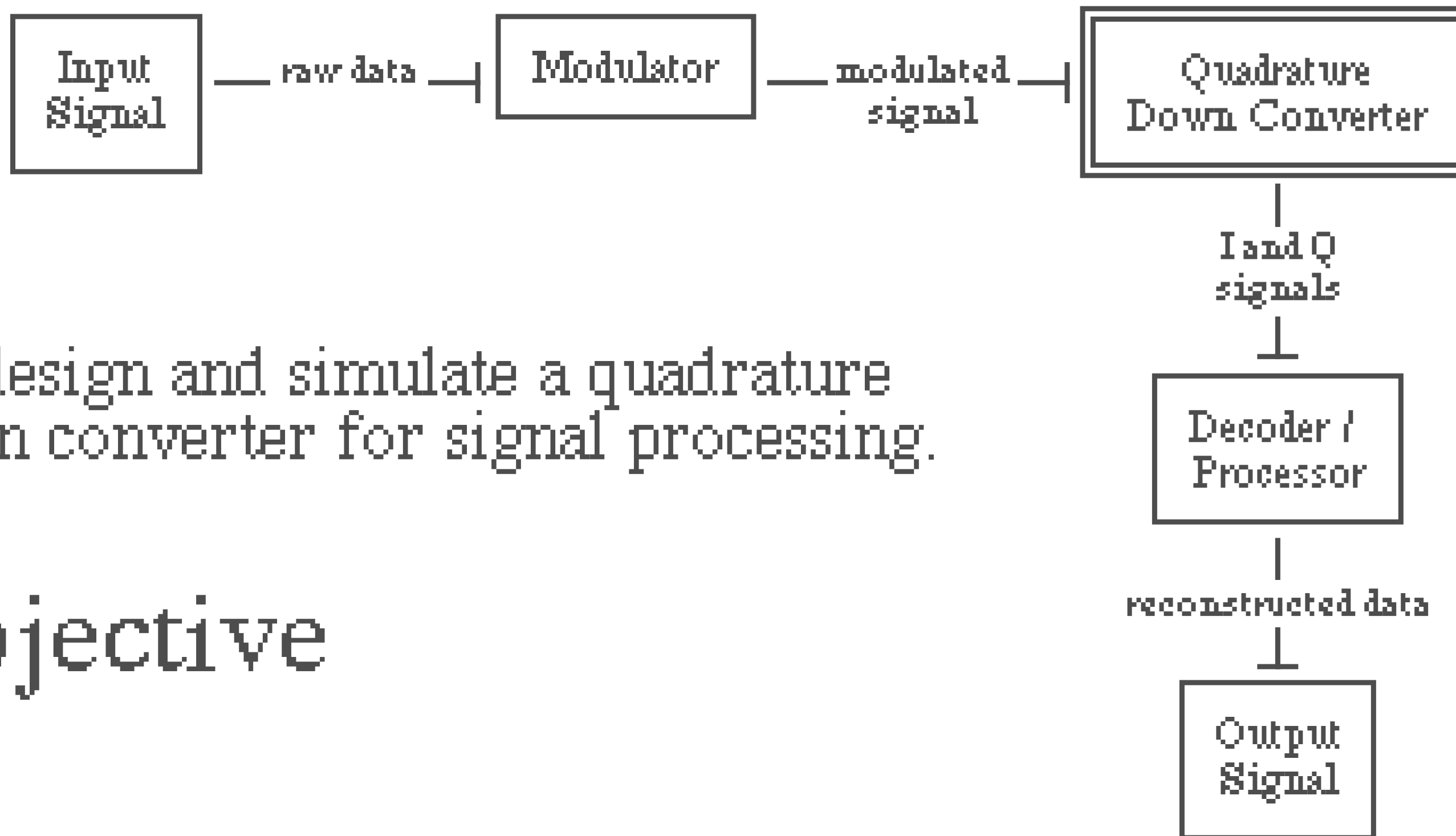
Project mid-evaluation

small enemy spider

Harry Jain 2024112005

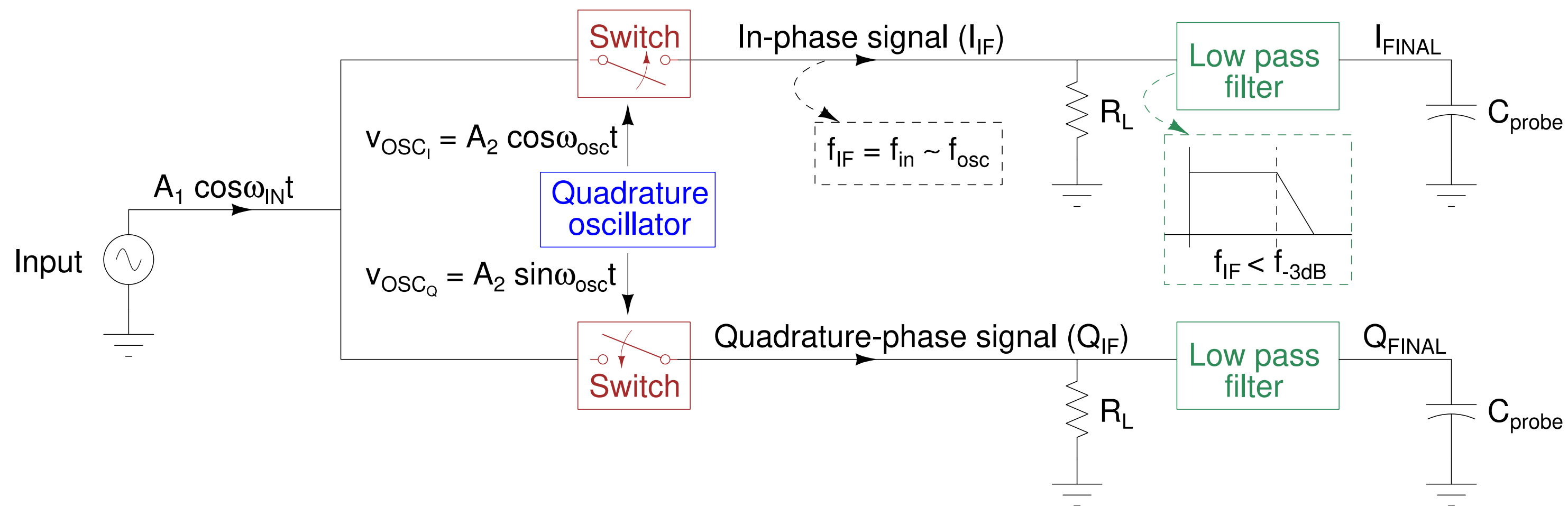
Kshitij Patkar 2024112001

Sanshray Vinesh 2024102071



To design and simulate a quadrature down converter for signal processing.

Objective



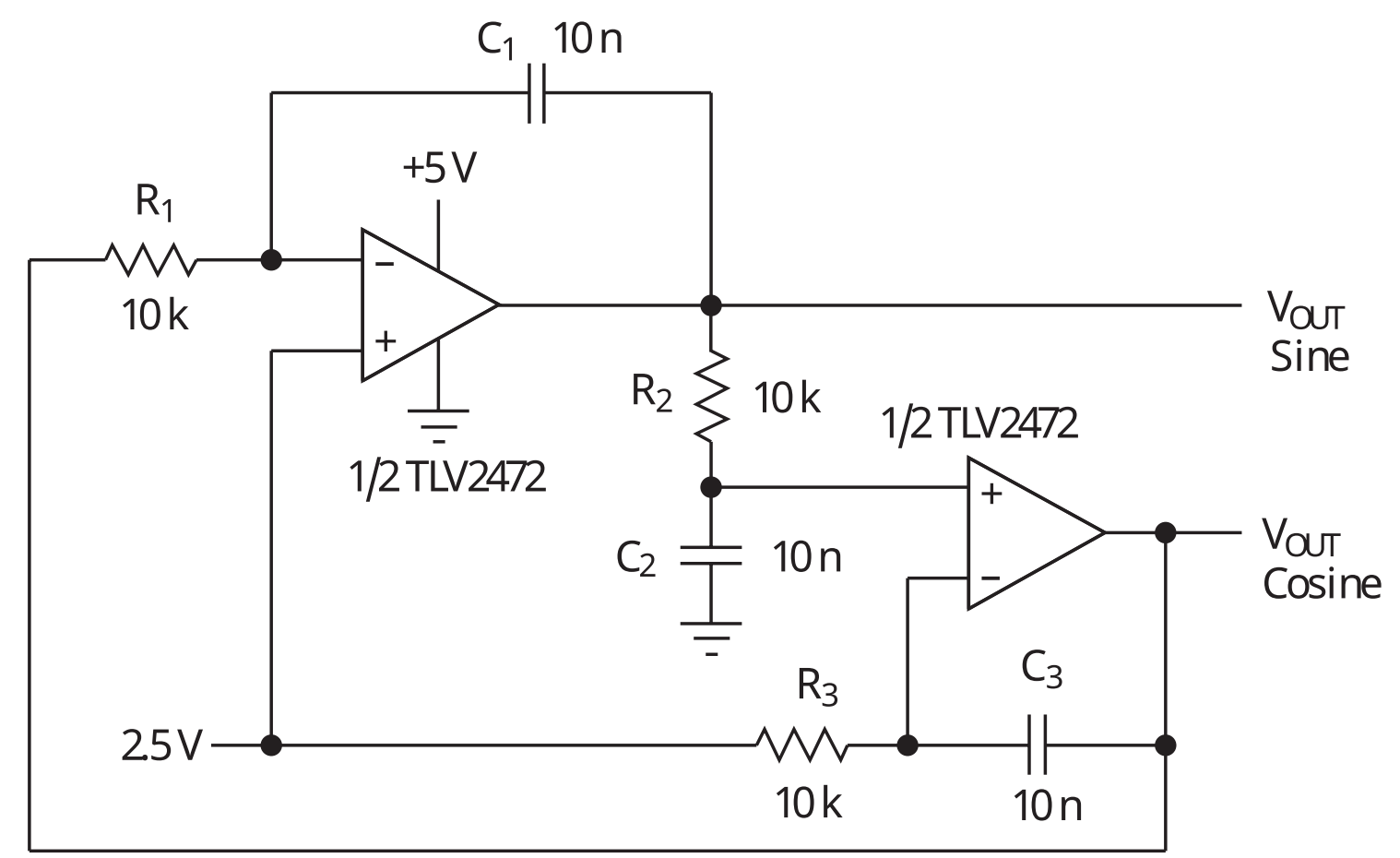
Block Diagram



Quadrature Oscillator

Generates two sinusoidal signals
with a phase difference of 90 deg.

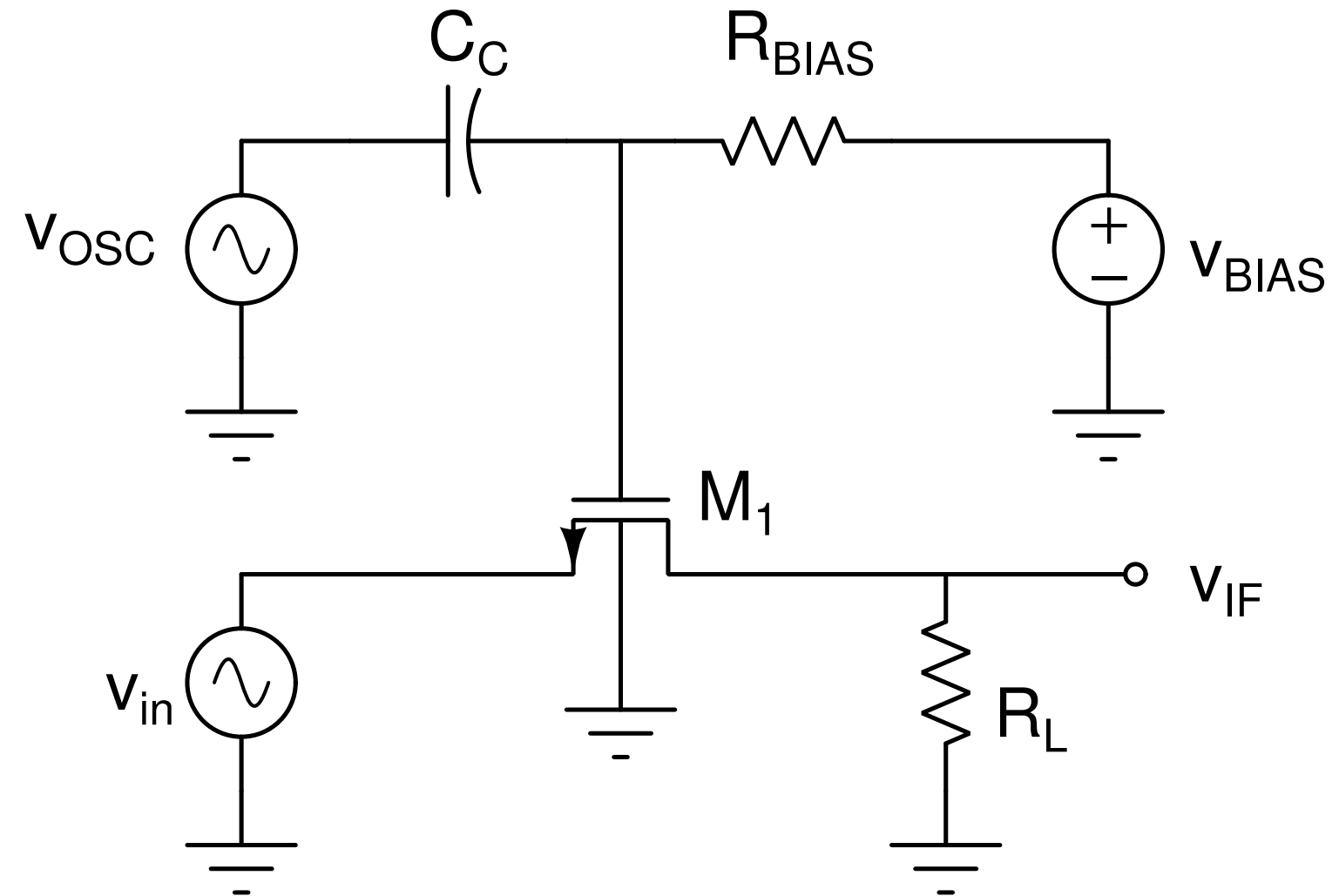
Local Oscillator

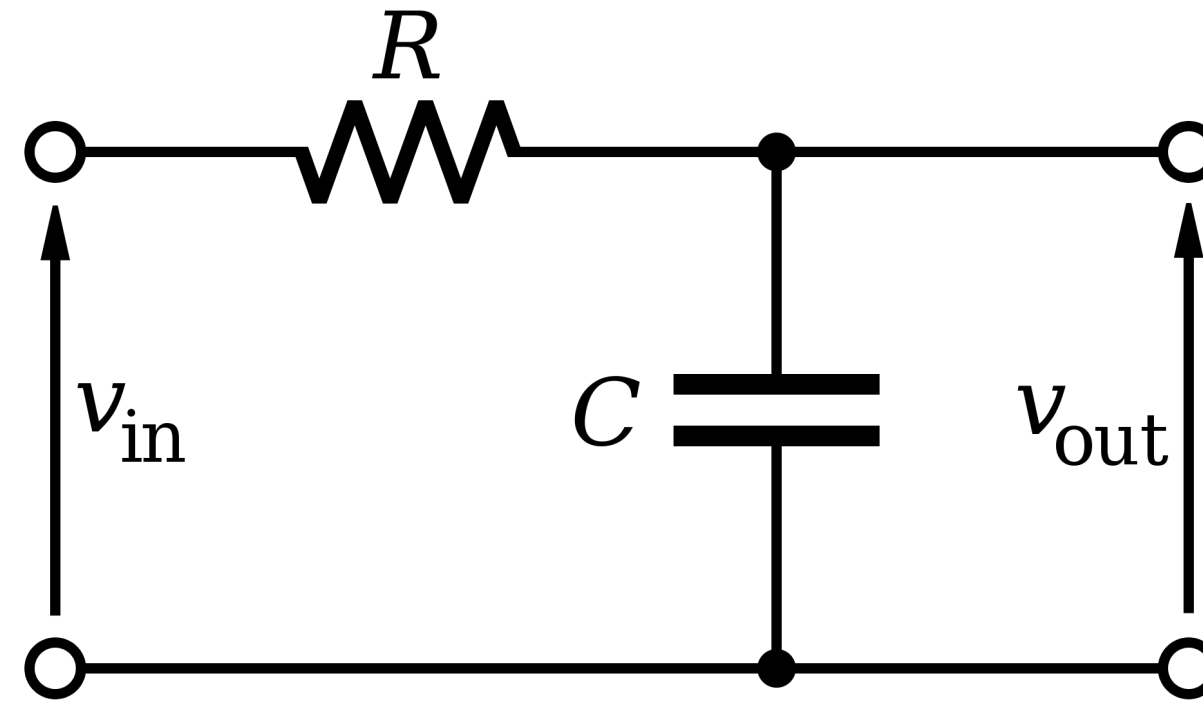


Multiplier

Multiplies the input signal with the biased oscillator signals

Mixer (Switcher)





Stops the junk higher
frequencies from passing

Low Pass Filter